

Discussion 11 Solutions

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Formula Bank

- **Probability:** For an event A in a sample space S , $P(A) = \frac{|A|}{|S|}$
 - **Conditional Probability:** The probability of A given B , $P(A | B) = \frac{P(A \cap B)}{P(B)}$
 - **Law of Total Probability:** If $\{B_1, B_2, \dots, B_n\}$ is a partition of the sample space, $P(A) = \sum_{i=1}^n P(A | B_i)P(B_i)$
 - **Bayes' Rule:** $P(B_i | A) = \frac{P(A|B_i)P(B_i)}{P(A)} = \frac{P(A|B_i)P(B_i)}{\sum_{j=1}^n P(A|B_j)P(B_j)}$
 - **Expectation:** For a discrete random variable X , $\mathbb{E}[X] = \sum_x x P(X = x)$
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Probability

1. A committee of 5 is to be selected from a group of 6 men and 9 women. If the selection is made randomly, what is the probability that the committee consists of 3 men and 2 women?

Solution:

The total number of committees is

$$\binom{15}{5}.$$

The number of committees with 3 men and 2 women is (choose 3 of the 6 men and 2 of the 9 women)

$$\binom{6}{3} \binom{9}{2}.$$

Thus the probability is

$$P = \frac{\binom{6}{3} \binom{9}{2}}{\binom{15}{5}} = \frac{720}{3003} \approx 0.2397.$$

$P \approx 0.2397$

2. A 5-card poker hand is said to be a full house if it consists of 3 cards of the same denomination and 2 other cards of the same denomination (of course, different from the first denomination). Thus, one kind of full house is three of a kind plus a pair. What is the probability that one is dealt a full house?

Solution: Total number of 5-card hands:

$$\binom{52}{5}$$

Number of full house hands (three of a kind + a pair):

- Choose the rank for the three of a kind: $\binom{13}{1}$
- Choose 3 suits out of 4 for that rank: $\binom{4}{3}$
- Choose the rank for the pair: $\binom{12}{1}$
- Choose 2 suits out of 4 for that rank: $\binom{4}{2}$

So the total number of full houses is

$$\binom{13}{1} \binom{4}{3} \binom{12}{1} \binom{4}{2} = 13 \cdot 4 \cdot 12 \cdot 6 = 3744.$$

Therefore,

$$P(\text{full house}) = \frac{\# \text{ full houses}}{\# \text{ 5-card hands}} = \frac{3744}{\binom{52}{5}} = \frac{3744}{2,598,960} \approx 0.00144.$$

So the probability of being dealt a full house is about 0.144%.

3. An elementary school is offering 3 language classes: one in Spanish, one in French, and one in German. The classes are open to any of the 100 students in the school. There are 28 students in the Spanish class, 26 in the French class, and 16 in the German class. There are 12 students that are in both Spanish and French, 4 that are in both Spanish and German, and 6 that are in both French and German. In addition, there are 2 students taking all 3 classes.
- (a) If a student is chosen randomly, what is the probability that the student is not in any of the language classes?
- (b) If a student is chosen randomly, what is the probability that the student is taking exactly one language class?

Solution

We have three sets: Spanish (S), French (F), and German (G).

All 3 classes: Students in $S \cap F \cap G = 2$

Exactly 2 classes:

$$\text{S and F only: } 12 - 2 = 10$$

$$\text{S and G only: } 4 - 2 = 2$$

$$\text{F and G only: } 6 - 2 = 4$$

Exactly 1 class:

$$\text{Spanish only: } 28 - 10 - 2 - 2 = 14$$

$$\text{French only: } 26 - 10 - 4 - 2 = 10$$

$$\text{German only: } 16 - 2 - 4 - 2 = 8$$

Students in no classes:

$$\text{Total in at least one class: } 14 + 10 + 8 + 10 + 2 + 4 + 2 = 50$$

$$\text{Students in no classes: } 100 - 50 = 50$$

(a)

$$P(\text{not in any class}) = \frac{50}{100} = \frac{1}{2} = 0.5$$

(b) Students in exactly one class = $14 + 10 + 8 = 32$

$$P(\text{exactly one class}) = \frac{32}{100} = \frac{8}{25} = 0.32$$

Conditional probability:

1. In Section 1, problem 3, what is the probability that a student selected at random is taking French, given that the student is also taking Spanish?

Solution:

We need to find $P(\text{French} \mid \text{Spanish})$.

$$P(F \mid S) = \frac{P(F \cap S)}{P(S)}$$

Given Information:

- Total students: 100
- Students taking Spanish: 28
- Students taking both Spanish and French: 12

$$P(\text{French} \mid \text{Spanish}) = \frac{\text{Students in both S and F}}{\text{Students in S}} = \frac{12}{28} = \frac{3}{7} \approx 0.4286$$

2. A family has two children. It is known that at least one of them is a boy. What is the probability that the other child is a girl?

Solution:

The two children have a sample space given by

$$\{(B, B), (B, G), (G, B), (G, G)\},$$

each with probability $1/4$.

If we let B be the event that at least one child is a boy and E be the event that one child is a girl and one child is a boy, the probability that we want to compute is

$$P(E \mid B) = \frac{P(E \cap B)}{P(B)}.$$

Now

$$P(B) = \frac{3}{4},$$

and $E \cap B$ consists of the set $\{(B, G), (G, B)\}$, so

$$P(E \cap B) = \frac{1}{2}.$$

Thus,

$$P(E \mid B) = \frac{1/2}{3/4} = \frac{2}{3}.$$

3. A laboratory blood test is 95 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a “false positive” result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then, with probability .01, the test result will imply that he or she has the disease.) If .5 percent of the population actually has the disease, what is the probability that a person has the disease given that the test result is positive?

Solution. Let D be the event that the person tested has the disease and E the event that the test result is positive.

Given:

- $P(E \mid D) = 0.95$ (test correctly detects the disease)
- $P(E \mid D^c) = 0.01$ (false positive rate)
- $P(D) = 0.005$ (0.5% of the population has the disease)

- $P(D^c) = 1 - P(D) = 0.995$

Then the desired probability is

$$\begin{aligned}
 P(D \mid E) &= \frac{P(D \cap E)}{P(E)} \\
 &= \frac{P(E \mid D)P(D)}{P(E \mid D)P(D) + P(E \mid D^c)P(D^c)} \\
 &= \frac{(.95)(.005)}{(.95)(.005) + (.01)(.995)} \\
 &= \frac{95}{294} \approx .323.
 \end{aligned}$$

Expectation:

1. A restaurant runs a promotion: every customer draws one card from a shuffled deck of 10 cards.
 - 2 cards give a 50% discount,
 - 3 cards give a 20% discount,
 - 5 cards give no discount.

Emily's meal costs \$48 before any discount. Compute the expected amount she pays.

Solution:

Let X be the amount Emily pays.

Possible discounts and corresponding payments:

$$\begin{aligned}
 \text{50\% discount: } X &= 48 \times 0.5 = 24, \\
 \text{20\% discount: } X &= 48 \times 0.8 = 38.4, \\
 \text{No discount: } X &= 48.
 \end{aligned}$$

Probabilities:

$$P(50\%) = \frac{2}{10}, \quad P(20\%) = \frac{3}{10}, \quad P(\text{none}) = \frac{5}{10}.$$

Therefore, the expected amount she pays is

$$\begin{aligned}\mathbb{E}[X] &= 24 \left(\frac{2}{10} \right) + 38.4 \left(\frac{3}{10} \right) + 48 \left(\frac{5}{10} \right) \\ &= 4.8 + 11.52 + 24 \\ &= 40.32.\end{aligned}$$

$$\boxed{\mathbb{E}[X] = \$40.32}$$

2. A child asks his parents for some money. The parents make the following offers.

- Father's offer: The child rolls a fair die. He will pay the child \$4 times the number rolled.
- Mother's offer: The child flips a fair coin. If it lands heads, she pays the child \$22. If it lands tails, she pays nothing.

Whose offer should the child consider?

Solution:

Father's offer: The child rolls a fair die. The payment is $4 \times (\text{number rolled})$. Since the die is fair, every outcome 1, 2, 3, 4, 5, 6 has probability $\frac{1}{6}$.

The expected value of a fair die roll is

$$\mathbb{E}[\text{roll}] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5.$$

So the father's expected payment is

$$\mathbb{E}[F] = 4 \times 3.5 = 14.$$

Mother's offer: Flip a fair coin:

$$\text{Heads} \Rightarrow \$22, \quad \text{Tails} \Rightarrow \$0.$$

Thus,

$$\mathbb{E}[M] = \frac{1}{2}(22) + \frac{1}{2}(0) = 11.$$

Conclusion:

$$\mathbb{E}[F] = \$14 \quad \text{and} \quad \mathbb{E}[M] = \$11.$$

$$\boxed{\text{The child should consider the father's offer.}}$$